



Team Details

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Maverick

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1	Team Leader	Mandeep Singh Rawat	Graduated
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i Team of 2 (Mandeep Singh Rawat, Vinay Pandey).

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Problem Statement Addressed



Drift-Sense: Navigation-Error Recovery for Wafer Inspection Tools — Applied Materials, Track 2

DESCRIPTION / DETAILS

Inspection tools revisit the same die site thousands of times a day at nanometre accuracy. Stage drift — thermal, vibration, mechanical slack — makes each revisit land several pixels off. Every die repeats the same layout, so the wrong site looks almost identical to the right one. Template matching fails exactly where needed: DRAM arrays and FinFET fin grids. Cost: corrupted process-control data, wasted tool time, scrapped wafers.



Idea Description – Describe your Idea/Solution/Prototype



Provide a brief summary of your idea, including the key concept and approach. Provide a brief overview of your solution and how it addresses the problem statement.

KEY CONCEPT & APPROACH

Both DRAM-style and FinFET-style dies, solved by one pipeline.

Classical, CPU-only, training-free: coarse-to-fine NCC over scale and rotation.

Key insight: the lattice says WHAT the pattern is; across-die process variation and micron-scale landmarks say WHERE.

The same pipeline also runs on 3-channel optical microscope captures, unmodified.

SOLUTION OVERVIEW

Physics-grounded SEM and optical dataset generators with independent noise per capture.

Inference takes two image paths, prints one sub-pixel (x, y) .

93.33 % within 5 px (112/120 pairs, four seeds); median error 0.09 px; ~2.3 s/pair on CPU.



Proposed Solution – The dataset generator



Describe your idea in detail. Include the methodology, technologies involved, and how it addresses the chosen problem statement.

GENERATOR

*GENERATOR — one 8000×8000 procedural layout per pair, so both captures share real structure.
DRAM: word/bit-lines + contact vias. FinFET: dense fins + periodic gate rows.
Realism: per-feature CD and overlay jitter; via open/short and fin-break defects.
Across-die CD non-uniformity field, amplitude randomized per pair (domain randomization).
Die composition: sub-array mats, independent preset and phase each, separated by periphery strips.
Micron-scale landmarks (0.9–2.6 μm): overlay marks, landing pads, CMP dummy-fill, coarse logic.
Two independent SEM captures: edge brightening $\rightarrow$ probe blur $\rightarrow$ Poisson shot noise $\rightarrow$ read noise.
Search image at lower dose (noisier, softer); $\pm 1.5^\circ$ rotation, $\pm 3\%$ scale jitter.
OPTICAL (bonus) — thin-film interference colour, per-channel diffraction-limited PSF $\rightarrow$ 3-channel RGB.*



Proposed Solution – The localisation pipeline



Describe your idea in detail. Include the methodology, technologies involved, and how it addresses the chosen problem statement.

PIPELINE

*PIPELINE – denoise, then coarse-to-fine NCC over scale ($\pm 12\%$) and rotation ($\pm 2^\circ$).
Denoise first: mild Gaussian blur kills independent sensor noise, keeps layout structure.
Scale ratio $10\times$ for a 1-channel SEM pair, $3\times$ for 3-channel optical – read from channel count.
Extract ALL near-equal peaks: in a periodic array, multi-peak ambiguity is the normal case.
Low-pass below array pitch to isolate the process-variation fingerprint; re-score candidates.
Colour pairs correlate all three channels jointly – in optical the fingerprint lives in hue.
Fingerprint overrides the lattice winner only on strong evidence – a confident lattice win is never demoted.
Nearest-to-centre rule for true ties, then parabolic sub-pixel fit $\rightarrow (x, y)$.*



Innovation and Uniqueness



Highlight what makes your idea unique or innovative compared to existing solutions.

KEY INNOVATION

Periodicity-suppressed verification: lattice is high-frequency, across-die variation is low-frequency — low-pass filtering separates them and breaks ties pure NCC cannot.

Domain randomization: a fixed strong variation field inflated our accuracy to 100 %, so we randomized it and report the honest number.

COMPETITIVE ADVANTAGE

Template matching assumes one global maximum, so it fails on periodic arrays.

Training-free, CPU-only, deterministic — no GPU, no weights, no retraining.

Sub-pixel accuracy (0.09 px median) at ~2.3 s per pair.



Measurement and Validation



How we know the numbers hold: per-layer ablations, a shipped honesty flag, and a contract test suite.

PER-LAYER ABLATIONS

Layered die realism, each layer measured: one uniform array 40 % under fingerprint ablation; mats + strips 90 %; micron landmarks 95 %.

Honest ablation shipped: the `--visual-clarity` flag makes cleaner images by disabling both cues — it costs 24 points (71.2 %), off by default.

Fingerprint fully disabled (`--pv-amplitude 0.0`, seed 404): still 19/20, median 0.10 px — the non-periodic micron landmarks carry it alone.

Scale and rotation edges also hold: 9:1 and 11:1 magnification ratio both 100% (20/20); 2° rotation edge 100% (20/20) — no measurable cost outside the nominal 10:1 / 1.5° defaults.

COVERAGE AND GUARDRAILS

One pipeline covers DRAM and FinFET with no per-architecture retuning.

Also covers SEM and 3-channel optical capture: the localiser reads the channel count and adapts.

28 contract tests, including one that fails loudly if the optical path degenerates into `applyColorMap`.



Impact and Benefits



Explain how your solution will make an impact, such as improving performance, reducing costs, increasing efficiency, or solving other challenges.

Primary Impact

Restores navigation on the periodic layouts where classical matching fails.
Fewer failed revisits, rescans and manual re-registrations on costly tools.
No GPU and no training — deployable in existing tool software, no hardware change.

Quantifiable Outcomes

Threshold-wise (112/120 pairs, four seeds): $\leq 5\text{px}$ 93.33% — $\leq 4\text{px}$ 93.33% — $\leq 2\text{px}$ 90.83% — $\leq 1\text{px}$ (sub-pixel) 90.0%.
Median localization error 0.09 px — sub-pixel.
Error stats: mean 16.1 px, worst-case 721 px — rare full lattice jumps, not near-misses.
100 % (20/20) at both 2× and 3× dose reduction — noise is not the limiting factor.
~2.3 s per 1000×1000 pair, one CPU core.
Optical microscope bonus: 8/8 within 5 px, median 0.55 px.
Honest limits: failures concentrate on near-uniform FinFET dies; boundary-straddling crops drop to 85 %; the clean-image build to 71.2 %.



Technology & Feasibility/Methodology Used



Describe the technologies, methodologies, or tools you plan to use to implement your idea.

IMPLEMENTATION STRATEGY

Stack: Python 3.13 + NumPy + OpenCV + Gradio. No DL framework, no weights, no training step.

Hardware: a standard laptop CPU. No GPU, no cloud platform.

Generation ~2 s/pair; inference ~2.3 s per 1000×1000 pair, single-threaded, deterministic.

Reproducible: seeded RNG, pinned requirements.txt, 28-test pytest suite, README runs clone → generate → localize → evaluate.

Classical signal processing throughout — auditable and portable to a tool controller.



generate_dataset.py
localize.py — (x, y) out
evaluate.py — accuracy
app.py — browser demo
tests/ — 28 tests



Commodity x86-64 CPU
No GPU required
~1 GB RAM



NumPy, OpenCV, Gradio
Git / GitHub
Windows + Linux tested



GitHub & Video Link



GitHub Repository

<https://github.com/Mandy434/drift-sense>



Prototype / Simulation Video

Not included – optional per problem statement; core deliverables (code, results, PPT) stand on their own.



Research Background



Research Background & Methodology

Briefly describe the research foundation or scientific principles supporting your idea.

*Every noise model and augmentation follows published SEM physics and process-variation literature.
SEM chain: secondary-electron edge contrast, Gaussian probe PSF, dose-dependent Poisson statistics.
Structure: DRAM word-line/bit-line/via organisation; FinFET fin-plus-gate cell geometry.
Process variation: systematic across-die CD non-uniformity — the physical reason a site is findable at all.
Optical modality: thin-film interference reflectance, diffraction-limited imaging, chromatic aberration, Bayer demosaic correlation.
Airy disc approximated as a Gaussian, $\sigma \approx 0.21 \lambda/\text{NA}$ — the Rayleigh radius as sigma over-blurs by $\sim 1.5\times$.
Solvability: a margin bug let ~ 1 pair in 30 be unsolvable by construction; found, fixed, and now a regression test.
Multi-seed reporting: at $n=20$ the standard error is ~ 5 pp, so one run cannot separate two variants.
Full choice-by-choice citation map is in references.md in the repository.*



References

References & Citations

List key papers, articles, or data sources.

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